

BTCUSDT 90-to-30 Future-Regime Dataset Generator

Corrected BXP Edition - Beginner Manual



The chart uses Matplotlib Axes.bxp, named colours green and red, and a white border around the plotting area.

1. What the generator creates

Each supervised example contains one 90-candle input image and the cluster ID of the immediately following 30-candle window.

```
input: candles t..t+89
target label: cluster ID of candles t+90..t+119
```

The windows are adjacent and do not overlap. The target chart is used only to calculate the label and must never be supplied to the supervised classifier.

2. Correct image folders

The corrected package separates the two image types:

images/inputs_90/ contains only the 90-candle images used for supervised training.

images/targets_30/ contains the 30-candle audit images used by ResNet-18, PCA and K-Means to assign target_cluster_id.

3. Before running

Fully extract the ZIP. Keep all files together. Install Python 3.12 64-bit; Python 3.14 is not recommended for this package.

You need an hourly BTCUSDT CSV containing timestamp, open, high, low and close columns.

4. Windows workflow

1. Double-click install_requirements.bat. It creates a private Python 3.12 environment named .venv and installs the packages.
2. Open config.json. Replace the path values required for your computer. The supplied path strings have otherwise been preserved.
3. Double-click run_generator.bat.
4. When generation finishes, double-click validate_dataset.bat.

Do not begin supervised training unless validation ends with All checks passed.

5. Configuration that should remain fixed

```
"window_size": 90
"forecast_horizon": 30
"stride": 30
```

The chart colours are the literal names green and red. The plot_padding_fraction value of 0.10 keeps the white border around the image. The extra_embargo_candles value of 120 strengthens split-boundary protection.

6. Leakage protections

Windows are created in chronological order. A target starts exactly one candle after its input ends. Samples that would cross split boundaries are removed. An additional 120-candle embargo is applied. PCA and K-Means are loaded as frozen models and are not refitted. The test split must be used only after model selection.

7. Validation checks

The validator confirms that every input contains 90 candles, every target contains 30 candles, the windows do not overlap, input paths point only to images/inputs_90, target paths point only to images/targets_30, all

referenced files exist, and train/validation/test boundaries do not share raw candles.

8. What to share after generation

For review, share manifests/dataset_summary.json, train_samples.csv, validation_samples.csv, test_samples.csv, duplicate_images.json, skipped_samples.txt and the final validation output. You do not initially need to upload all chart images.

Cluster mapping

Cluster ID	Interpretive name
0	High-volatility consolidation
1	Bullish trend and breakout
2	Volatile sideways and reversal
3	Bearish trend and breakdown